A+B

BET-C610
SEMESTER EXAMINATION- MAY 2024
CLASS: B.TECH SEMESTER:VI
EMBEDDED SYSTEMS

Time: 3 hours **Max. Marks: 70**

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)		CO	BL
Instructions: Answer any ***five*** questions in about 150 words each. Each question carries six marks. **(5 X 6 = 30 Marks)**			
Q-1	Explain the concept of hardware and software partitioning in embedded system design.	CO5	L6
Q-2	Explain monolithic and layered structure of operating system.	CO1	L1
Q-3	Explain the design life cycle of embedded systems. Identify and describe the key stages involved in the development process, from conception to deployment.	CO6	L6
Q-4	Discuss the characteristics and applications of open collector outputs, I/O sinking and sourcing techniques in embedded systems.	CO3	L3
Q-5	Discuss the Timer/Counter in auto reload mode (Mode 2) of 8051 Microcontroller.	CO2	L2
Q-6	Write an assembly program for 8051 to copy a block of 10 bytes of data from RAM locations starting at 35H to RAM locations starting at 60H.	CO2	L2
Q-7	Discuss the role of Real-Time Operating Systems (RTOS) in embedded systems. Explain the tasks, states, semaphores, and shared data management in RTOS.	CO1	L1
Q-8	Explain the concept of Harvard and Von Neumann architectures in microcontrollers. Compare their advantages and disadvantages in embedded system design.	CO3	L3
Q-9	What is mailbox, message queue and message pipe?	CO1	L1
Q-10	Explain the role of communication protocols in embedded systems. Compare and contrast serial protocols, parallel protocols, and wireless protocols. Provide examples of each protocol type.	CO2	L2

SECTION-B (LONG ANSWER TYPE QUESTIONS) **Instructions**: Answer any ***four*** questions in detail. Each question carries 10 marks. **(4 X 10 = Marks)**		CO	BL
Q-1	Analyze the design parameters of an embedded system and discuss their significance in system development. How do these parameters influence the functionality and efficiency of embedded systems?	CO4	L4
Q-2	Compare and contrast General Purpose Processors with Application Specific Instruction set Programming (ASIPs). Discuss their respective features, advantages, and applications in embedded system development.	CO1	L1
Q-3	Analyze the architecture of 8051 microcontrollers. Discuss its assembly language, registers, addressing modes, and instruction set. How do interrupts, timer/counter, and serial communication enhance its functionality?	CO2	L2
Q-4	Draw the interfacing circuit and write a program in C language to interface Common Cathode Seven segment LED with microcontroller 8051.	CO3	L3
Q-5	Draw the interfacing circuit and write a program in C language to interface Stepper motor with microcontroller 8051.	CO3	L3
Q-6	Explain the importance of optimizing programs in embedded systems. Discuss Finite State Machine with Data Path (FSMD) and its role in program optimization. Provide examples of program optimization techniques.	CO1	L1
Q-7	Write short notes of following 1. ADC(Analog o Digital Converter) 2. Push Buttons interfacing with 8051 controller.	CO2	L2
Q-8	Compare and contrast the features of 80386, 80486, and ARM processors.	CO2	L2

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